Praesidium

Neighborhood Safety Simulation & Planning — Simulation Report

Generated 10/15/2026, 2:54:14 AM

Project: Vasanth Nagar East

Vasanth Nagar East

A fictional mid-size Indian neighborhood used for demonstrations: two main roads, three cross lanes, a small bazaar street, a lakeside park, a bus stop, and a community centre. Comes with street lighting, a few CCTV cameras, and two night patrol guards. Fully editable.

Important: this report presents outputs of a simplified simulation with synthetic agents and fictional scenario events. It is a planning-support exercise — NOT a prediction of actual crime, a photometric assessment, or a guarantee of safety. Modeled coverage does not establish that incidents would be prevented, recorded, or detected.

Data provenance & run identity

Data source	Bundled demonstration neighborhood (fictional geometry)
Engine version	1.0.0
Simulation seed	20260214
Simulated duration	30 simulated minutes
Scenario	Typical Night — A typical night with normal modeled pedestrian activity, routine patrols, and a low rate of fictional incidents.
Synthetic population	48 agents (65% residents / 35% night-shift commuters)
Baseline run ID	run_mv8maxwo1z3
Intervention run ID	run_mv8maxxj2moe

Scenario configuration & assumptions

Start time (display)	22:00
Expected simulated events	6 per simulated hour (Poisson model)
Lighting outage assumption	4% of lamps unavailable this night
Ambient visibility assumption	100% (multiplier on modeled illumination)
CCTV effective range multiplier	1.00×
Pedestrian walking speed	1.35 m/s (±15% per agent)
Guards	As defined in the neighborhood
Guard walking speed	1.15 m/s

Methodology: pedestrian agents route along the walkable network using A* pathfinding. Modeled lighting coverage samples the walkable network every 5 m and sums each active lamp's brightness × (1 " d/radi as lit above a threshold that rises as ambient visibility falls. Modeled CCTV coverage requires a sample point to lie within a camera's range × range multiplier and horizontal field of view. Guard response times are estimated from simulated travel distance along the network at the configured guard speed. Event generation uses a seeded Poisson-process model; event types and severities are fictional scenario parameters.

Baseline results

Metric	Value	Label
Modeled lit path coverage	44.1%	modeled coverage
Modeled CCTV coverage	2.4%	modeled coverage
Poorly lit path segments	8 of 10	modeled
Lighting outages this night	5	scenario assumption
Trips with reachable destinations	89.3%	simulated
Unreachable trips	8	simulated
Average route length	258.8 m	simulated
Route completion rate	95.5%	simulated
Simulated events	7	fictional scenario events
Events with an available guard	100.0%	simulated
Events reached by a guard	100.0%	simulated
Estimated avg. response time	194 s	estimated
Estimated median response time	193 s	estimated
Unreachable events	0	simulated
Unresolved events	2	simulated
Resolution rate	71.4%	simulated
Patrol coverage (40 m of waypoints)	41.0%	modeled

Simulated events by type (baseline)

Event type (fictional scenario)	Count
Nuisance	2

Mugging	2
Medical assistance	1
Vandalism	1
Harassment	1

Notable unresolved cases (baseline)

Time	Type	Severi	Outcome	Detail
96s	Mugging	sev 3	Escaped in a blind spot	Synthetic mugger incident · in a CCTV blind
983s	Harassment	sev 2	Escaped in a blind spot	Synthetic harasser incident · in a CCTV blind

Intervention results

Metric	Value	Label
Modeled lit path coverage	44.3%	modeled coverage
Modeled CCTV coverage	6.2%	modeled coverage
Poorly lit path segments	8 of 10	modeled
Lighting outages this night	5	scenario assumption
Trips with reachable destinations	89.3%	simulated
Unreachable trips	8	simulated
Average route length	258.8 m	simulated
Route completion rate	95.5%	simulated
Simulated events	7	fictional scenario events
Events with an available guard	100.0%	simulated
Events reached by a guard	100.0%	simulated
Estimated avg. response time	145 s	estimated
Estimated median response time	168 s	estimated
Unreachable events	0	simulated
Unresolved events	2	simulated
Resolution rate	71.4%	simulated
Patrol coverage (40 m of waypoints)	41.0%	modeled

Baseline vs intervention

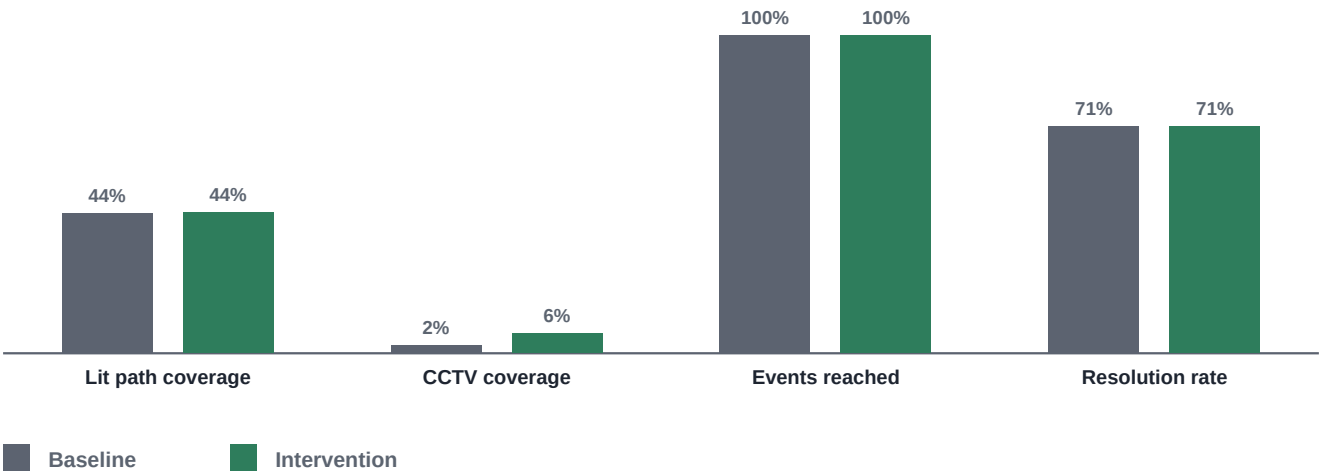
Metric	Baseline	Intervention	Change
Modeled lit path coverage	44.1%	44.3%	+0.2 pp
Modeled CCTV coverage	2.4%	6.2%	+3.8 pp
Poorly lit path segments	8	8	no change
Estimated avg. guard response	194 s	145 s	-49 s
Estimated median guard response	193 s	168 s	-25 s
Events reached by a guard	100.0%	100.0%	no change
Resolution rate (simulated)	71.4%	71.4%	no change
Unresolved simulated events	2	2	no change
Trips with reachable destinations	89.3%	89.3%	no change
Route completion rate	95.5%	95.5%	no change
Patrol coverage (40 m of waypoints)	41.0%	41.0%	no change
Average route length	259 m	259 m	no change
Perpetrator incidents (simulated)	3	3	no change
Incidents detected	100.0%	100.0%	no change
Incidents detected by CCTV	0.0%	66.7%	+66.7 pp

Incidents in CCTV blind spots	3	1	" 2
Perpetrators caught	33.3%	33.3%	no change
Median detection delay	3 s	3 s	no change
Median guard response (detection to scene)	157 s	165 s	+8 s
Guard time asleep (min)	7.3	10.3	+3

Estimated intervention cost: INR 53,400

Cost assumptions (estimates only): streetlight INR 14,000 · CCTV camera INR 26,000 · guard-night INR 3,200 · lamp relocation/edit "H 10% of install. Costs exclude tax, civil works and recurring maintenance.

Key modeled-coverage metrics at a glance



Implemented interventions

- Edited 1 streetlight (position, brightness, radius or status).
- Added 2 CCTV cameras.

For the committee: what to fund first

SIMULATED ESTIMATES

Same night, one change — Mugging at 22:01: escaped in blind spot -> caught on CCTV (traceable)

We replayed the same simulated night twice with identical people and perpetrators (seed 20260214). With all proposed changes (est. INR 53,400), the risk index moves from 6.1 to 3.5 points (2.6 points lower); 1 incident(s) end with the perpetrator caught, and 2 fewer incident(s) happen in camera blind spots. To see which purchase does the work, each change was then tested on its own against the original night.

#	Change	Est. cost	Risk	Per lakh	Avoided	Caught	Faster
1	New CCTV camera at (-144, 72)	INR 26,000	-1.7	6.5	-	+1	104 s
2	New CCTV camera at (-7, -4)	INR 26,000	-1.6	6.2	-	-	-
3	Move / adjust streetlight sl_x	INR 1,400	-	-	-	-	-
All	All changes together	INR 53,400	-2.6	4.9	-	-	-8 s

- Best value: "New CCTV camera at (-144, 72)" — 6.5 risk points removed per INR 1 lakh.
- 1 paid change(s) showed no measurable benefit on this night: Move / adjust streetlight sl_x. Consider deferring or re-siting them.
- Changes interact: single-change effects need not add up to the combined result.
- One simulated night is one sample. Re-test on the High-Risk and Low-Visibility nights before committing funds.

How to read it: "Risk" is the change in a simple risk index (lower is better). Risk index = sum of incident severity (1-3) × outcome weight (escaped in blind spot 1.0 | no guard 0.9 | unreachable 0.85 | still open 0.8 | escaped but on CCTV 0.5 | caught 0.15). "Per lakh" = risk points removed per INR 1,00,000 spent. "Avoided" = incidents that did not occur; "Caught" = extra incidents ending with the perpetrator caught; "Faster" = median guard response time saved. Costs are standard estimates excluding tax, civil works and maintenance.

Limitations & responsible use

- All simulated residents, guards and events are synthetic and fictional. No real individual is profiled, tracked, or scored.
- Event generation is a fictional scenario parameter, not an inference about actual criminality of any area or person.
- Modeled lighting and CCTV coverage are geometric approximations with disclosed assumptions; they are not photometric or engineering assessments.
- Camera coverage does not imply incidents would be detected or recorded; lighting coverage does not imply prevention.
- Response times are estimates from simulated distances along the modeled pedestrian network at configured speeds.
- Results depend on scenario parameters, synthetic population assumptions and geometry quality. Validate findings with real community consultation.
- If the neighborhood was imported from OpenStreetMap, geometry is simplified and attributes may be inferred or missing.
- This report is generated locally in your browser. No project data leaves your device except optional AI planning requests, which send only an aggregated summary.